

Sharan Giri

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PROFESSIONAL STATEMENT

2 years shipping production automation pipelines at Barclays for Markets and Private Banking under SLA-bound delivery and live production support. Now building end-to-end LLM and RAG systems including multi-turn agentic pipelines and hybrid retrieval, with a focus on turning raw data into reliable, explainable decisions, deployed on GCP and Kubernetes.

EDUCATION

Khoury College of Computer Science, Northeastern University M.S. in Data Science (GPA: 4.0) Secured Third Place at Google-Northeastern Expo for AudioSeek (Dec 2025)	Boston, MA Sep 2024 – May 2026
Sri Sivasubramaniya Nadar College of Engineering, Anna University B.E. in Computer Science Engineering (GPA: 3.7)	India Aug 2018 – Jun 2022

EXPERIENCE

Barclays Global Service Centre – Markets And Private Banking Technology Software & Automation Engineer Data Pipeline and API Integration - Built production-grade pipelines parsing unstructured email and API/JSON sources into structured formats, improving throughput by 50% and saving 85 hours/month - Enforced error volume thresholds and automated stakeholder alerting across 6 production workflows, reducing false-positive alerts by 72% and improving system reliability - Migrated 4 UI-based automations to API-driven components, cutting turnaround by 33% and eliminating manual dependencies across Markets and Private Banking - Owned SLA-bound production support with a 2-day resolution target, debugging live failures, coordinating controlled rollbacks and redeploying after regression testing	Aug 2022 – Jun 2024 India
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PROJECTS

CourseLENS Governed Multi-Turn AI Reasoning System <i>SIGDIAL 2026 (Under Review)</i> - Built a Socratic tutoring system guiding students through C++ concepts via retrieved course evidence, not direct answers - Architected a 4-stage Socratic FSM that enforces guided reasoning through deterministic state control, not prompting - Designed a 90/10 hybrid retrieval layer confining responses to top-5 verified course chunks, eliminating hallucination risk - Validated 15/16 adversarial compliance checks and 96.15% answer faithfulness, ensuring pedagogical integrity at scale	Jan 2026 – Apr 2026
AudioSEEK Production RAG Platform <i>Google-Northeastern Expo</i> - Built a RAG platform converting 30+ hours of audio into a searchable knowledge base with answers under 15 seconds - Validated transcription across 3 models using WER and ROUGE-L thresholds, selecting the best output per audiobook - Engineered dual FAISS and GCP vector stores with chapter-isolated indexes to prevent out-of-window content surfacing - Deployed containerized GKE microservices via Terraform with MLflow tracking latency, token use, and query drift	Sep 2025 – Dec 2025
FinAgent Agentic vs. RAG Evaluation System SEC Financial Analysis - Designed a controlled experiment comparing an agentic pipeline to baseline RAG on identical LLMs and schemas - Ingested 72 SEC 10-K filings across 16 companies, normalizing 5-year financial records for controlled peer comparison - Evaluated outputs on evidence coverage, consistency, and usefulness to measure agentic pipeline gains over naive retrieval - Enforced schema validation across all LLM outputs, achieving 95%+ faithfulness on 1,500+ structured signals from filings	Mar 2026 – Apr 2026
Early Product Success Prediction Demand Signal Modeling and Feature Engineering - Built a PySpark ETL pipeline converting raw Amazon Beauty reviews into a scored dataset for 28-day launch prediction - Engineered 40 demand signals across rating trends, sentiment, and review velocity to surface early commercial traction - Trained multimodal XGBoost models under A/B ablation, achieving Lift@100 of 2.6x and Precision@100 of 0.65 - Delivered SHAP-backed explainability so category managers can identify winning launches and pull back weak ones early	Nov 2025 – Dec 2025

TECHNICAL SKILLS

Software & Data Engineering: Python, SQL, REST APIs, FastAPI, PySpark, Pandas, Airflow, Git, CI/CD (GitHub Actions)
AI Systems & Retrieval: RAG Pipelines (Retrieval Augmented Generation), Agentic AI Systems, Multi-Agent Orchestration, LangChain, LangGraph, FAISS, Vector Databases, BM25, BGE-M3, HuggingFace, Pydantic, Prompt Engineering, Structured Output Generation
ML & Modeling: PyTorch, Scikit-learn, XGBoost, Feature Engineering, SHAP, Ragas, LLM Evaluation, ML Observability
MLOps & Infrastructure: Docker, Kubernetes (GKE), MLflow, Terraform
Cloud Platforms: GCP (Cloud Run, GCS, Firestore, Cloud SQL), Gemini Flash